Established – 1961 Subject: web

SEVA SADAN'S

R. K. TALREJA COLLEGE

OF

ARTS, SCIENCE & COMMERCE

ULHASNAGAR – 421 003

CERTIFICATE

This is to certify that Mr. Anurag Manoj Yadav **of F.Y. Information Technology (FYIT) Roll No.** 2541057 **has satisfactorily completed the Web Designing Mini Project entitled -** Portfolio with JavaScript Logic Based Utility Feature **during the academic year 2025 – 2026, as a part of the practical requirement. The project work is found to be satisfactory and is approved for submission.**

PROF. INCHARGE **HEAD OF DEPT**

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Topic - Portfolio with JavaScript Logic Based Utility Feature

1. INTRODUCTION

This project is a responsive portfolio website integrated with JavaScript logic-based utility features to make it more interactive and user-friendly. Unlike a basic static website, this portfolio uses JavaScript to add dynamic behavior such as scroll position indication, theme switching, and form validation. The main aim of the project is to combine design skills with programming logic. It demonstrates how frontend technologies can work together to create a professional and modern web experience.

The website is developed using HTML for structure, CSS for styling, and JavaScript for logical functionality. JavaScript is used to handle events like scrolling, button clicks, and form submissions. For example, the scroll position indicator calculates how much of the page has been viewed and updates a progress bar accordingly. These logic-based features improve user engagement and make the website visually informative.

The implementation focuses on DOM manipulation and event handling concepts in JavaScript. Conditional statements and functions are used to validate user input in the contact form and to control UI changes like dark and light mode. The website is tested on different browsers to ensure responsiveness and smooth performance. The final project is hosted using **GitHub** and deployed through **GitHub Pages**.

In conclusion, this project successfully integrates web design with logical programming concepts. It enhances the user experience through interactive utility features and dynamic content behavior. The project also strengthens understanding of JavaScript fundamentals such as events, functions, and calculations. Overall, it serves as a strong

foundation for developing advanced and fully functional web applications in the future.

2. REQUIREMENT SPECIFICATION

Requirement specification describes the resources and tools needed to develop and run the project successfully. For the **Portfolio with JavaScript Logic-Based Utility Feature**, both hardware and software requirements are minimal, making the project simple to develop, test, and deploy. The system is lightweight and does not require high-end configuration, which makes it accessible for students and beginners

2.1 Hardware Requirements

The following hardware components are required to develop and run the project:

- Computer or Laptop
- Minimum 4 GB RAM
- Keyboard and Mouse
- Internet connection (for testing, hosting, and reference)

2.2 Software Requirements

The software requirements for this project are:

- Operating System: Windows 10 or above
- Web Browser: Google Chrome, Mozilla Firefox, or Microsoft Edge
- Code Editor: Visual Studio Code / Notepad++ / Any Text Editor
- Version Control & Hosting: **GitHub**
- Deployment Platform: **GitHub Pages**
- Technologies Used:
    - HTML5
    - CSS3
    - JavaScript

2.3 Functional Requirements

The system should provide the following functionalities:

- Responsive Navigation Bar
- Scroll Position Indicator

• BMI Calculator with Input Validation
• Dynamic Result Display using JavaScript
• Contact Form Section
• Smooth Scrolling Between Sections

2.4 Non-Functional Requirements

The system should ensure:
• Simple and clean user interface
• Fast loading time
• Browser compatibility
• Easy navigation
• Good visual appearance
• Mobile responsiveness

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3. SYSTEM DESIGN

System design defines the overall structure and working flow of the Portfolio with JavaScript Logic-Based Utility Feature project. The system follows a simple three-layer architecture consisting of structure, presentation, and logic layers. This modular approach ensures that each part of the website functions independently while working together to provide a smooth user experience.

3.1 System Architecture

The project is based on a front-end architecture that includes:

• **HTML (Structure Layer)** – Defines the layout of sections such as Home, About, Skills, Projects, BMI Calculator, and Contact.
• **CSS (Presentation Layer)** – Provides styling, layout design, responsiveness, colors, fonts, and animations.
• **JavaScript (Logic Layer)** – Implements dynamic functionality such as BMI calculation, input validation, scroll behavior, and interactive elements.

3.2 Module Design

The system is divided into the following modules:

• Navigation Module – Handles menu links and smooth scrolling between sections.
• Hero & About Module – Displays personal information and introduction.
• Skills & Projects Module – Shows technical skills with progress bars and project details.
• BMI Calculator Module – Performs logical calculations using JavaScript and displays categorized results.
• Contact Module – Collects user information through a contact form interface.

3.3 Data Flow Design

• User enters height and weight in the BMI Calculator.
• JavaScript validates the input values.
• BMI formula is applied to calculate the result.
• The calculated BMI value and health category are displayed dynamically on the screen.

3.4 User Interface Design

The user interface is designed to be clean, modern, and responsive. A consistent color theme is used throughout the website. Sections are clearly separated for better readability and navigation. The layout adapts to different screen sizes, ensuring proper display on desktops, tablets, and mobile devices.

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4. SYSTEM IMPLEMENTATION

System implementation describes how the designed system is developed using selected technologies and tools. The Portfolio with JavaScript Logic-Based Utility Feature is implemented using HTML, CSS, and JavaScript. Each component of the website is coded separately and then integrated to ensure smooth functionality. The project follows a structured file organization with separate files for HTML (structure), CSS (design), and JavaScript (logic).

4.1 Frontend Implementation

• HTML is used to create the basic structure of the website including sections like Home, About, Skills, Projects, BMI Calculator, and Contact.
• CSS is used to design the layout, colors, fonts, spacing, responsiveness, and animations.
• Media queries are implemented to make the website mobile-friendly and responsive across different devices.

4.2 JavaScript Logic Implementation

• JavaScript functions are created to calculate BMI using the standard formula:
BMI = Weight (kg) / Height (m^2).
• Input validation is implemented to ensure users enter valid height and weight values.
• Conditional statements are used to categorize BMI results (Underweight, Normal, Overweight, Obese).
• DOM manipulation is used to dynamically display calculated results without refreshing the page.

4.3 Integration and Deployment

• All files (HTML, CSS, JavaScript, images) are integrated within a single project folder.
• The project is tested in modern browsers to ensure proper performance.
• The website is uploaded to **GitHub** for version control and management.
• Finally, it is deployed using **GitHub Pages** to make it accessible online.

The system implementation ensures that the website runs efficiently, provides interactive features, and delivers a smooth user experience.

Here is the code used to create Portfolio

HTML

```
    <!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Anurag Yadav | Portfolio</title>
  <link rel="stylesheet" href="style.css">
  <!-- Font Awesome for Icons -->
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.4.0/css/all.min.css">
</head>
<body>

  <!-- Navigation -->
  <nav class="navbar">
    <div class="nav-container">
      <a href="#home" class="logo">Anurag<span>Yadav</span></a>
      <ul class="nav-menu">
        <li><a href="#home" class="nav-link">Home</a></li>
        <li><a href="#about" class="nav-link">About</a></li>
        <li><a href="#skills" class="nav-link">Skills</a></li>
        <li><a href="#projects" class="nav-link">Projects</a></li>
        <li><a href="#bmi-calculator" class="nav-link">BMI Calculator</a></li>
```

```
            <li><a href="#contact" class="nav-link">Contact</a></li>
        </ul>
        <div class="hamburger">
            <span class="bar"></span>
            <span class="bar"></span>
            <span class="bar"></span>
        </div>
    </div>
</nav>

<!-- Hero Section -->
<section id="home" class="hero">
    <div class="hero-container">
        <div class="hero-content">
            <h3>Hello, It's Me</h3>
            <h1>Anurag Yadav</h1>
            <h3>And I'm a <span class="typing-text">First Year Student</span></h3>
            <p>Welcome to my portfolio! I'm a passionate first-year student exploring the world
of programming and web development.</p>
            <div class="social-icons">
                <a href="#" class="social-link"><i class="fab fa-github"></i></a>
                <a href="#" class="social-link"><i class="fab fa-linkedin-in"></i></a>
                <a href="#" class="social-link"><i class="fab fa-instagram"></i></a>
                <a href="#" class="social-link"><i class="fab fa-twitter"></i></a>
            </div>
            <a href="#about" class="btn">More About Me</a>
        </div>
        <div class="hero-image">
            <div class="image-wrapper">
                <img src="anu.png" alt="Anurag Yadav">
            </div>
        </div>
    </div>
</section>

<!-- About Section -->
<section id="about" class="about">
    <div class="about-container">
        <h2 class="section-title">About <span>Me</span></h2>
        <div class="about-content">
            <div class="about-image">
                <img src="anu.png" alt="About Anurag">
            </div>
            <div class="about-text">
                <h3>First Year Computer Science Student</h3>
                <p>Hi! I'm Anurag Yadav, a first-year student passionate about technology and
programming. I'm currently learning various programming languages and web development
technologies. My journey in the world of coding has just begun, and I'm excited to explore,
learn, and create amazing projects.</p>
```

```
                <p>I believe in continuous learning and am dedicated to building a strong
foundation in computer science. When I'm not coding, you can find me exploring new
technologies, contributing to open-source projects, or improving my problem-solving
skills.</p>

                <div class="about-info">
                  <div class="info-box">
                    <span>Name:</span>
                    <p>Anurag Yadav</p>
                  </div>
                  <div class="info-box">
                    <span>Age:</span>
                    <p>18-20 Years</p>
                  </div>
                  <div class="info-box">
                    <span>Qualification:</span>
                    <p>First Year Student</p>
                  </div>
                  <div class="info-box">
                    <span>Nationality:</span>
                    <p>Indian</p>
                  </div>
                </div>
              </div>
            </div>
          </div>
        </section>

        <!-- Skills Section -->
        <section id="skills" class="skills">
          <div class="skills-container">
            <h2 class="section-title">My <span>Skills</span></h2>
            <div class="skills-grid">
              <div class="skill-card">
                <div class="skill-icon">
                  <i class="fab fa-html5"></i>
                </div>
                <h3>HTML5</h3>
                <div class="progress-bar">
                  <div class="progress" style="width: 85%"></div>
                </div>
                <span class="skill-level">85%</span>
              </div>
              <div class="skill-card">
                <div class="skill-icon">
                  <i class="fab fa-css3-alt"></i>
                </div>
                <h3>CSS3</h3>
                <div class="progress-bar">
                  <div class="progress" style="width: 80%"></div>
```

```
                </div>
                <span class="skill-level">80%</span>
            </div>
            <div class="skill-card">
                <div class="skill-icon">
                    <i class="fab fa-js"></i>
                </div>
                <h3>JavaScript</h3>
                <div class="progress-bar">
                    <div class="progress" style="width: 70%"></div>
                </div>
                <span class="skill-level">70%</span>
            </div>
            <div class="skill-card">
                <div class="skill-icon">
                    <i class="fas fa-code"></i>
                </div>
                <h3>C++</h3>
                <div class="progress-bar">
                    <div class="progress" style="width: 75%"></div>
                </div>
                <span class="skill-level">75%</span>
            </div>
        </div>
    </div>
</section>

<!-- Projects Section -->
<section id="projects" class="projects">
    <div class="projects-container">
        <h2 class="section-title">My <span>Projects</span></h2>
        <div class="projects-grid">
            <div class="project-card">
                <div class="project-icon">
                    <i class="fas fa-calculator"></i>
                </div>
                <h3>BMI Calculator</h3>
                <p>A functional Body Mass Index calculator that helps users calculate their BMI and
understand their health status. Built with HTML, CSS, and JavaScript.</p>
                <div class="project-tags">
                    <span>HTML</span>
                    <span>CSS</span>
                    <span>JavaScript</span>
                </div>
                <a href="#bmi-calculator" class="project-btn">View Project</a>
            </div>
            <div class="project-card">
                <div class="project-icon">
                    <i class="fas fa-user-tie"></i>
                </div>
```

```
                <h3>Portfolio Website</h3>
                <p>A personal portfolio website showcasing my skills, projects, and achievements.
Fully responsive and professionally designed.</p>
                <div class="project-tags">
                  <span>HTML</span>
                  <span>CSS</span>
                </div>
                <a href="#home" class="project-btn">View Project</a>
              </div>
              <div class="project-card coming-soon">
                <div class="project-icon">
                  <i class="fas fa-code-branch"></i>
                </div>
                <h3>More Coming Soon</h3>
                <p>I'm constantly learning and working on new projects. Stay tuned for more
exciting additions to my portfolio!</p>
                <div class="project-tags">
                  <span>Coming Soon</span>
                </div>
                <a href="#" class="project-btn disabled">View Project</a>
              </div>
            </div>
          </div>
        </section>

        <!-- BMI Calculator Section -->
        <section id="bmi-calculator" class="bmi-calculator">
          <div class="bmi-container">
            <h2 class="section-title">BMI <span>Calculator</span></h2>
            <p class="bmi-description">Use this calculator to check your Body Mass Index (BMI).
This calculator is built using JavaScript as requested!</p>

            <div class="bmi-wrapper">
              <div class="bmi-form">
                <h3>Calculate Your BMI</h3>
                <div class="input-group">
                  <label for="height">Height (cm)</label>
                  <input type="number" id="height" placeholder="Enter your height in cm"
min="1" max="300">
                </div>
                <div class="input-group">
                  <label for="weight">Weight (kg)</label>
                  <input type="number" id="weight" placeholder="Enter your weight in kg"
min="1" max="500">
                </div>
                <button class="btn calculate-btn" onclick="calculateBMI()">Calculate BMI</button>
                <button class="btn reset-btn" onclick="resetBMI()">Reset</button>
              </div>

              <div class="bmi-result">
```

```
                <h3>Your Result</h3>
                <div id="result-display">
                  <p class="placeholder-text">Enter your height and weight above, then click
"Calculate BMI"</p>
                </div>
                <div class="bmi-categories">
                  <div class="category underweight">
                    <span>Underweight</span>
                    <p>&lt; 18.5</p>
                  </div>
                  <div class="category normal">
                    <span>Normal</span>
                    <p>18.5 - 24.9</p>
                  </div>
                  <div class="category overweight">
                    <span>Overweight</span>
                    <p>25 - 29.9</p>
                  </div>
                  <div class="category obese">
                    <span>Obese</span>
                    <p>&gt; 30</p>
                  </div>
                </div>
              </div>
            </div>
          </div>
        </section>

        <!-- Contact Section -->
        <section id="contact" class="contact">
          <div class="contact-container">
            <h2 class="section-title">Contact <span>Me</span></h2>
            <div class="contact-content">
              <div class="contact-info">
                <h3>Get In Touch</h3>
                <p>Feel free to reach out to me for any collaboration, project ideas, or just to say
hello!</p>
                <div class="contact-details">
                  <div class="contact-item">
                    <i class="fas fa-envelope"></i>
                    <span>anuragyadav@email.com</span>
                  </div>
                  <div class="contact-item">
                    <i class="fas fa-phone"></i>
                    <span>+91 98765 43210</span>
                  </div>
                  <div class="contact-item">
                    <i class="fas fa-map-marker-alt"></i>
                    <span>India</span>
                  </div>
```

```
                </div>
                <div class="social-icons">
                    <a href="#" class="social-link"><i class="fab fa-github"></i></a>
                    <a href="#" class="social-link"><i class="fab fa-linkedin-in"></i></a>
                    <a href="#" class="social-link"><i class="fab fa-instagram"></i></a>
                    <a href="#" class="social-link"><i class="fab fa-twitter"></i></a>
                </div>
            </div>
            <form class="contact-form" action="#" method="POST">
                <div class="form-group">
                    <input type="text" name="name" placeholder="Your Name" required>
                </div>
                <div class="form-group">
                    <input type="email" name="email" placeholder="Your Email" required>
                </div>
                <div class="form-group">
                    <input type="text" name="subject" placeholder="Subject" required>
                </div>
                <div class="form-group">
                    <textarea name="message" placeholder="Your Message" rows="5"
required></textarea>
                </div>
                <button type="submit" class="btn">Send Message</button>
            </form>
        </div>
    </div>
</section>

<!-- Footer -->
<footer class="footer">
    <div class="footer-container">
        <p>Copyright &copy; 2024 Anurag Yadav | All Rights Reserved</p>
        <div class="footer-links">
            <a href="#home">Home</a>
            <a href="#about">About</a>
            <a href="#contact">Contact</a>
        </div>
    </div>
</footer>

<script src="javascript.js"></script>

</body>
</html>
```

Javascript

```
function calculateBMI() {
```

```
// Get input values
const heightInput = document.getElementById('height').value;
const weightInput = document.getElementById('weight').value;
const resultDisplay = document.getElementById('result-display');

// Validate inputs
if (heightInput === '' || weightInput === '') {
    resultDisplay.innerHTML = '<p class="error-text">Please enter both height and weight!</p>';
    return;
}

if (heightInput <= 0 || weightInput <= 0) {
    resultDisplay.innerHTML = '<p class="error-text">Please enter valid positive numbers!</p>';
    return;
}

// Calculate BMI
const heightInMeters = heightInput / 100;
const bmi = weightInput / (heightInMeters * heightInMeters);
const bmiFixed = bmi.toFixed(1);

// Determine category
let category = '';
let categoryClass = '';

if (bmi < 18.5) {
    category = 'Underweight';
    categoryClass = 'underweight';
} else if (bmi >= 18.5 && bmi < 25) {
    category = 'Normal Weight';
    categoryClass = 'normal';
} else if (bmi >= 25 && bmi < 30) {
    category = 'Overweight';
```

```
            categoryClass = 'overweight';
        } else {
            category = 'Obese';
            categoryClass = 'obese';
        }

        // Display result
        resultDisplay.innerHTML = `
            <div class="bmi-value ${categoryClass}">
                <span class="value">${bmiFixed}</span>
                <span class="label">Your BMI</span>
            </div>
            <div class="bmi-category ${categoryClass}">
                <span class="category-label">Category:</span>
                <span class="category-value">${category}</span>
            </div>
            <p class="bmi-message">${getBMIMessage(category)}</p>
        `;
    }

    function getBMIMessage(category) {
        switch(category) {
            case 'Underweight':
                return 'You may need to gain some weight. Consider consulting a healthcare provider for personalized advice.';
            case 'Normal Weight':
                return 'Great! Your weight is in the healthy range. Keep maintaining your healthy lifestyle!';
            case 'Overweight':
                return 'You may need to lose some weight for better health. Consider regular exercise and a balanced diet.';
            case 'Obese':
                return 'Your BMI indicates obesity. Please consult a healthcare provider for guidance on achieving a healthier weight.';
            default:
```

```
            return '';
        }
    }

    function resetBMI() {
        document.getElementById('height').value = '';
        document.getElementById('weight').value = '';
        document.getElementById('result-display').innerHTML = '<p class="placeholder-text">Enter your height and weight above, then click "Calculate BMI"</p>';
    }
```

CSS

```
/* ================= GOOGLE FONT ================= */
@import url('https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600;700&display=swap');

*{
   margin:0;
   padding:0;
   box-sizing:border-box;
   font-family:'Poppins',sans-serif;
   scroll-behavior:smooth;
}

body{
   background:#0f172a;
   color:#fff;
}

section{
   padding:100px 10%;
}

.section-title{
```

```
    text-align:center;
    font-size:42px;
    margin-bottom:60px;
}

.section-title span{
    color:#00f7ff;
}

/* ================= BUTTON ================= */
.btn{
    display:inline-block;
    padding:12px 30px;
    background:#00f7ff;
    color:#000;
    border-radius:40px;
    text-decoration:none;
    font-weight:500;
    transition:.3s ease;
    border:none;
    cursor:pointer;
}

.btn:hover{
    background:#fff;
}

/* ================= NAVBAR ================= */
.navbar{
    position:fixed;
    width:100%;
    top:0;
    left:0;
    background:#1e293b;
```

```
    z-index:1000;
}

.nav-container{
    display:flex;
    justify-content:space-between;
    align-items:center;
    padding:20px 10%;
}

.logo{
    font-size:24px;
    font-weight:600;
    color:#fff;
    text-decoration:none;
}

.logo span{
    color:#00f7ff;
}

.nav-menu{
    display:flex;
    list-style:none;
    gap:30px;
}

.nav-link{
    color:#fff;
    text-decoration:none;
    transition:.3s;
}

.nav-link:hover{
```

```
    color:#00f7ff;
}

/* ================= HERO ================= */
.hero{
    padding-top:150px;
}

.hero-container{
    display:flex;
    align-items:center;
    justify-content:space-between;
    gap:50px;
    flex-wrap:wrap;
}

.hero-content{
    flex:1;
}

.hero-content h1{
    font-size:48px;
}

.hero-content span{
    color:#00f7ff;
}

.hero-image{
    flex:1;
    text-align:center;
}

.hero-image img{
```

```
    width:350px;
    border-radius:50%;
    border:5px solid #00f7ff;
}

/* ================= ABOUT ================= */
.about-container{
    max-width:1200px;
    margin:auto;
}

.about-content{
    display:flex;
    gap:50px;
    align-items:center;
    flex-wrap:wrap;
}

.about-image img{
    width:350px;
    border-radius:10px;
}

.about-text{
    flex:1;
}

.about-info{
    margin-top:20px;
    display:grid;
    grid-template-columns:repeat(2,1fr);
    gap:15px;
}
```

```
.info-box span{
   color:#00f7ff;
   font-weight:600;
}

/* ================= SKILLS ================= */
.skills-grid{
   display:grid;
   grid-template-columns:repeat(auto-fit,minmax(250px,1fr));
   gap:30px;
}

.skill-card{
   background:#1e293b;
   padding:25px;
   border-radius:12px;
   text-align:center;
   transition:.3s;
}

.skill-card:hover{
   transform:translateY(-8px);
}

.skill-icon{
   font-size:45px;
   color:#00f7ff;
   margin-bottom:15px;
}

.progress-bar{
   background:#333;
   height:8px;
   border-radius:5px;
```

```
    margin:10px 0;
}

.progress{
    height:8px;
    background:#00f7ff;
    border-radius:5px;
}

/* ================= PROJECTS ================= */
.projects-grid{
    display:grid;
    grid-template-columns:repeat(auto-fit,minmax(280px,1fr));
    gap:30px;
}

.project-card{
    background:#1e293b;
    padding:30px;
    border-radius:12px;
    transition:.3s;
}

.project-card:hover{
    transform:scale(1.05);
}

.project-icon{
    font-size:40px;
    color:#00f7ff;
    margin-bottom:15px;
}

.project-tags span{
```

```
    background:#00f7ff;
    color:#000;
    padding:5px 12px;
    border-radius:20px;
    font-size:12px;
    margin-right:5px;
}

/* ================= BMI ================= */
.bmi-wrapper{
    display:flex;
    gap:40px;
    flex-wrap:wrap;
}

.bmi-form,
.bmi-result{
    flex:1;
    background:#1e293b;
    padding:30px;
    border-radius:12px;
}

.input-group{
    margin-bottom:20px;
}

input, textarea{
    width:100%;
    padding:12px;
    border:none;
    border-radius:8px;
    margin-top:8px;
}
```

```
.calculate-btn{
    margin-right:10px;
}

/* ================= CONTACT ================= */
.contact-content{
    display:flex;
    gap:50px;
    flex-wrap:wrap;
}

.contact-info,
.contact-form{
    flex:1;
    background:#1e293b;
    padding:30px;
    border-radius:12px;
}

.form-group{
    margin-bottom:20px;
}

.contact-item{
    margin-bottom:15px;
}

/* ================= FOOTER ================= */
.footer{
    background:#1e293b;
    padding:25px 10%;
    text-align:center;
}
```

```
.footer-links a{
   color:#00f7ff;
   margin:0 10px;
   text-decoration:none;
}
/* ================= RESPONSIVE ================= */
@media(max-width:768px){
   .hero-container,
   .about-content,
   .contact-content,
   .bmi-wrapper{
      flex-direction:column;
      text-align:center;
   }
   .nav-menu{
      display:none;
   }
}
```

---

5. SYSTEM TESTING AND RESULTS

System testing is performed to ensure that all features of the Portfolio with JavaScript Logic-Based Utility Feature work correctly and efficiently. The website was tested after completing the implementation phase to verify that each module performs as expected. Both functional and usability testing methods were applied to check overall performance and user interaction.

5.1 Functional Testing

• Navigation links were tested to ensure smooth scrolling between sections.
• The BMI Calculator was tested with different input values to verify correct calculation and category display.
• Input validation was checked to ensure error messages appear for empty or invalid entries.
• Contact form fields were tested to confirm proper input acceptance.

5.2 Compatibility Testing

- The website was tested on Google Chrome, Mozilla Firefox, and Microsoft Edge.
- Responsive design was tested on different screen sizes including desktop and mobile view.
- Layout and styling consistency were verified across different browsers.

5.3 Results

- All navigation links function correctly without errors.
- The BMI Calculator provides accurate results based on user input.
- The website loads quickly and performs smoothly.
- The design remains responsive and visually consistent on various devices.

6. FUTURE SCOPE AND CONCLUSION

Future Scope

The portfolio website can be enhanced further by adding more advanced features in the future. The design can be improved by introducing modern animations and interactive UI elements to make the website more attractive. Backend integration can be added to store user data, manage form submissions, and support authentication systems. More utility-based JavaScript features such as additional calculators, dynamic content loading, and real-time functionality can also be implemented. Future updates may focus on improving security, performance optimization, and overall user experience.

Conclusion

This project helped in understanding the practical implementation of web development technologies such as HTML, CSS, and JavaScript. It provided knowledge about designing responsive layouts and adding interactive logic-based features. The portfolio website successfully meets the basic project requirements by providing an attractive interface and functional utilities. Overall, this project was a valuable learning experience that improved both coding and design skills.

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7. REFERENCES

These references were used to understand web development concepts, HTML, CSS, and JavaScript implementation standards.

1. Mozilla Developer Network (MDN) – https://developer.mozilla.org/

2. W3Schools – https://www.w3schools.com/

3. World Wide Web Consortium (W3C) – https://www.w3.org/

8. GLOSSARY

HTML (HyperText Markup Language):
HTML is the standard markup language used to create the structure of web pages.
CSS (Cascading Style Sheets):
CSS is used to control the styling, layout, and visual presentation of web pages.
JavaScript:
JavaScript is a programming language used to add interactivity and dynamic functionality to websites.
Responsive Design:
Responsive design is an approach that makes web pages adapt to different screen sizes and devices.
Validation:
Validation is the process of checking user input to ensure it is correct and acceptable.
UI (User Interface):
UI refers to the visual elements of a website that allow users to interact with the system.
UX (User Experience):
UX focuses on improving user satisfaction by enhancing usability and accessibility of the website.